Drop2Crop — System User Manual

Smart Dual-Motor IoT Precision Irrigation & Fertigation Station

Target Device: ESP32-S3 (16MB Flash, 8MB PSRAM)	**Firmware Version:** v2.5-DualMotor-Production
Cloud Protocol: MQTT over Secure WebSockets (WSS)	**Broker:** broker.emqx.io (Port 8084 WSS / 1883 TCP)
Dual Actuators: Motor 1: Clean Water (Relay 1) | Motor 2: Fertigation (Relay 2)	**Timezone:** Asia/Dhaka (BST, UTC+6)
Dashboard URL: https://fahimashabb.github.io/drop2crop/	**Security Gateway:** 4-Digit Hardware-Verified PIN (Default: 2564)

1. System Overview & Dual-Motor Architecture

Drop2Crop is an advanced precision agricultural IoT station designed to deliver autonomous, data-driven crop irrigation and nutrient delivery without manual guesswork. The system features an industrial-grade ESP32-S3 microcontroller coupled to a 2-channel isolated relay board that operates two independent 12V pump circuits:

- **Motor 1 — Clean Water Pump (Relay 1 / GPIO 17 (Mirrored: GPIO 9)):** Draws fresh water from the main reservoir for normal hydration, moisture maintenance, and soil flushing.
- **Motor 2 — Fertigation Pump (Relay 2 / GPIO 18 (Mirrored: GPIO 10)):** Dedicated pump drawing liquid fertilizer, nutrient mix, or mineral additives from a separate fertilizer tank, preventing chemical fouling of the primary clean reservoir.
- **Independent & Coordinated Control:** Either pump can be run individually or both can be run simultaneously via the cloud web app hosted on GitHub Pages from anywhere across Bangladesh.

2. Zero-Code WiFi Provisioning & Relocation (Captive Portal)

Drop2Crop incorporates an intelligent **Captive Portal Provisioning System**. If you move the hardware to a new farm, switch routers, or use mobile hotspots, **you never need to plug the ESP32 into a computer or touch firmware code**:

- **Automatic Activation:** On boot, the ESP32 attempts to connect to saved WiFi. If disconnected for 12 seconds, it automatically broadcasts an open network: **Drop2Crop_WiFi_Setup**.
- **Manual Force Activation:** Hold the physical **BOOT button (GPIO 0)** on the ESP32 for 3 seconds while powered. The onboard LED will flash rapidly and launch the portal immediately.
- **Connect:** Open WiFi settings on your phone or PC, select **Drop2Crop_WiFi_Setup**.
- **Portal Page:** The browser automatically opens **http://192.168.4.1**. Select your 2.4 GHz WiFi network, enter the password, and click **Save & Connect**.
- **Instant Cloud Reconnection:** Credentials save to NVS flash, the ESP32 connects to the cloud, synchronizes Bangladesh Standard Time (UTC+6) via NTP, and resumes all operations.

3. Operational Modes & Multi-Motor Logic

Mode	Mode Name	Dual-Motor Behavior & Safeguards
Mode 0	**Smart Auto (Soil + Rain)**	Autonomous closed-loop irrigation. Turns **Motor 1 (Clean Water)** ON when soil moisture drops below **35%**, and shuts OFF at **70%** saturation. Both motors are automatically **inhibited for 24 hours** if rain occurs, and cut off if water level is below **15%**.
Mode 1	**Alert-Only Monitor**	Passive sensing. When soil drops below 35%, triggers audible cloud dashboard alarms and visual status warnings without running pumps. Farmer reviews condition and starts Water, Fertigation, or Both via manual buttons.
Mode 2	**Multi-Timer Schedule**	Clock-driven schedule. Each timer slot specifies target: ■ **Water Only**, ■ **Fertigation Only**, or ■ **Both Motors**. Executes at programmed times (e.g. 6:00 AM Water 20m, 6:00 PM Fertigation 15m).
Mode 3	**Manual Cloud Override**	Direct farmer control. Four one-touch buttons: *'Start Water Pump'*, *'Start Fert Pump'*, *'Start Both Motors'*, and *'Stop All Motors'*. Dry-run reservoir cutoff remains strictly active.

4. Precision Scheduling with Actuator Targeting

Drop2Crop's NTP hardware scheduler stores up to **10 distinct daily timer slots** permanently in NVS Flash. When adding a schedule slot, the farmer selects which motor will run:

- **Target 0 — Clean Water:** Ideal for routine morning hydration (e.g. 06:00 AM, 20 mins) or deep afternoon root soaking.
- **Target 1 — Fertigation:** Ideal for precise dosage delivery (e.g. Wednesday 08:00 AM, 10 mins) from the nutrient reservoir.
- **Target 2 — Both Motors:** Synchronous run for pre-mixed flood or combined drip lines.
- **Fail-Safe Clock:** Real-time clock tracks Bangladesh Standard Time via NTP. Continues executing offline even if Internet connection drops.
- **Anti-Duplicate Lockout:** A daily latch ensures schedules only run once per calendar day and reset cleanly at midnight (00:00:00).

5. Dual Sensor Safeguards & PIN-Protected Force Mode Bypass

24-Hour Rain Lockout: Raindrop detection starts an 86,400-second (24h) countdown that pauses automated & scheduled runs for both water and fertigation to prevent nutrient wash-out and waterlogging.

Ultrasonic Water Reservoir Monitoring: HC-SR04 sonar sensor measures depth from reservoir top:
• **Level < 20%:** Triggers amber dashboard warning to refill.
• **Level < 15%:** Hard emergency interlock automatically disengages both relays, preventing pump cavitation and motor burnout.

PIN-Protected Safety Mechanism Bypass (Force Mode): For bench testing, pipeline flushing, or emergency operation during low water readings, operators can toggle the safety mechanism OFF directly from the web dashboard. To prevent accidental pump damage or unauthorized tampering, toggling the safety mechanism requires entering the **Station Security PIN (default 2564)**. Once authorized, the system enters Force Mode, allowing Motor 1 and Motor 2 to be forcefully turned ON or OFF.

6. Device Online / Offline Status & Health Diagnostics

- **Real-Time Online/Offline Detection:** Heartbeat packets stream every 2 seconds. If power is lost or charger unplugged, dashboard alerts **'The Device is Not Online (Turned Off)'** within 6 seconds.
- **Red Power LED:** Solid red PWR LED illuminates upon 5V power connection.

- **WiFi Beacon:** If WiFi is missing, ESP32 broadcasts *'Drop2Crop_WiFi_Setup'*, proving the CPU and RF circuitry are 100% operational.
- **Relay Mechanical Clicks:** Testing 'Start Water Pump' or 'Start Fert Pump' produces audible relay clicks and lights individual red channel LEDs.

7. Access Security & Hardware Authentication

The web dashboard is protected by a 4-digit PIN lockscreen (default **2564**). Every MQTT command requires hardware authentication token verification on the ESP32-S3, preventing unauthorized remote access.